

Lab Assignments:Case study two

1) Consider the Leukemia dataset

- a) In example 3.2 the data has been analyzed by SMACOF method with the Euclidian distance. Please run the analysis once again but with Mahalanobis distance. Does the choice of distance lead to major change in results ?
- b) Analyze again the data by classical MDS method and compare the result with a) and also example 3.2

2) Consider the Stochastic Neighbor Embedding method

- a) Find the NIPS text data used in the paper of Hinton and Roweis [\[2002\]](#). Run both SNE and tSNE to analyze this dataset and see if tSNE does a better work.
- b) SNE has a lot of applications in various field, for instance, Autism Brain Imaging. SNE has been used to visualize group differences in functional connectivity in low-dimensional space. Please find a proper data set and analyze the data by SNE or tSNE.

3) In the original [paper](#) of Josh Tenenbaum et al, the authors illustrated the power of ISOMAP for face recognition. Please find the data set and run ISOMAP to reproduce the figure A.

4) **Optional:** Please listen to the [talk](#) of Prof Daubechies, where she mentioned the method [AriaDNE](#). What is AriaDNE ? Please find any data which could be better analyzed by AriaDNE.

On Wednesday Feb 11 , 2026, the time 14:15 – 15:00 is devoted to demonstration of solutions in classroom.

Three students will be randomly chosen to show their results.

Every student should also upload her/his report in Canvas at the latest 11:59 PM Feb 11, 2026.

Attendance is mandatory